

Basic structure of C++ program ÷

#include <iostream.h> ~~X~~ Modern compilers don't support it
standard

#include <iostream> input-output stream

Stream ÷ It is a sequence of bytes or characters.

→ This directive includes the <iostream> header file which contain the declaration for the input output stream classes and objects.

using namespace std; std namespace is the standard namespace in which all the C++ standard library functions and classes are defined.

```
int main ( )
```

```
{
```

```
    _____ statement ;
```

```
    _____ statement ;
```

```
    return 0;
```

```
}
```




Input & Output in C++

iostream ÷ standard input output stream

iostream is a header file in the C++ standard library that provides facilities for input and output through streams.

iostream library includes two primary types of streams ÷

- Input stream
- Output stream

Input stream ÷

istream is the base class for input streams in C++.

It provides the functionality to read data from input devices like keyboard.

Output stream ÷

ostream is the base class for output streams in C++.

It provides the functionality to write data to output devices like the console.

Input | Output objects defined in standard iostream library are

- cin → predefined object of class istream
- It is linked with standard input & keyboard
- cout → predefined object of class ostream
- It is linked with standard output & console

cerr

clog

input / output operators :-

>> Input operator

- It is known as **extraction** or **get from** operator
- It takes the value from the keyboard and assigns it to the right.

<< Output operator

- It is known as **injection** or **put to** operator
- It sends the contents of the variable to its left.

Output concept in C++ ÷



```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
    cout << "C++ is better than C.";
```

```
    ↓
```

It is a predefined object of class **ostream** used to output the text to the console.

```
    cout << "C++ is superset of C.";
```

New line followed by the text.

```
    cout << "C++ is superior than C." << endl;
```

endl is a manipulator that does two things.

1. Inserts a newline character '\n'.
2. Flushes the output stream to ensure that all buffered data is written out immediately. It means that any data that is currently buffered (i.e. temporarily stored in memory to be output later) is immediately written out to the intended output destination such as the console.

```
    cout << "C++ earlier known as C with Classes.";
```

```
    return 0;
```

```
}
```




Output Explanation :

- `cout << "C++ is better than C." ;`

This line outputs the text without adding a newline at the end.

- `cout << "\n C++ is superset of C." ;`

This line starts with a newline character '\n', which moves the cursor to the next line before outputting the text "C++ is superset of C."

- `cout << "\n C++ is superior than C." << endl ;`

This line starts with a newline character '\n', which moves the cursor to the next line before outputting the text.

The `<< endl` adds another newline character and flushes the output buffer.

- `cout << "C++ earlier known as C with classes" ;`

This line is printed on a new line because of the `endl` at the end of the third statement.

Input Concept in C++

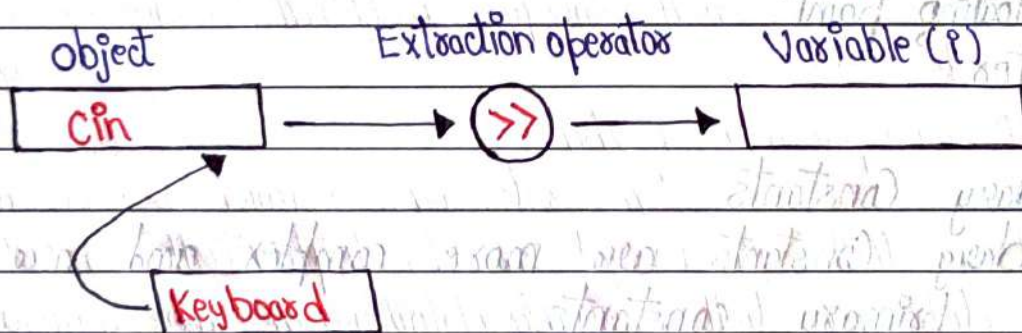
```
#include <iostream>
using namespace std;
int main ( )
{
```

```
    int i;
    cout << "Enter the value of i = ";
    cin >> i;
```



It is known as extraction operator

It will **take** the value from the **keyboard** and assigns it to the variable on its **right**.



```
    cout << "The value of i is : " << i << endl;
```

```
    return 0;
```

```
}
```

• Tokens ÷

The smallest individual units in a program are known as tokens.

- Constants
- Identifier
- Variable
- Keywords